Title

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INTRODUCTION

SETUP

Lorentzian Regge calculus

Regge calculus [1] is a discrete gravitational theory on simplicial manifolds Δ with the length of edges $\{l_e\}$ dynamical. In a Lorentzian setting, a simplicial manifold consists of intrinsically flat Lorentzian 4-simplices $\sigma \in \Delta$. Curvature is located at triangles $t \in \Delta$ and captured by deficit angles $\delta_t(\{l_e\})$, lying in the space orthogonal to t. For t spacelike (timelike), this space is isomorphic to $\mathbb{R}^{1,1}$ ($\mathbb{R}^2$).[1] Lorentzian deficit angles are defined as $\delta_t^{\mathrm{L},\pm} = \mp i2\pi - \sum_{\sigma\supset t}\psi_{\sigma,t}^{\mathrm{L},\pm}$, following the conventions of [2, 3], where the $\psi_{\sigma,t}$ are dihedral angles. "$\pm$" indicates a choice of sign which becomes important for causally irregular configurations [2–4] discussed below. The bulk action of Lorentzian Regge calculus is given by [1–3]

$$\begin{aligned} S_{\mathrm{R}}[\{l_e\}] = &\sum_{t\in\Delta:\ \mathrm{tl}} |A_t(\{l_e\})|\delta_t^{\mathrm{E}}(\{l_e\}) \\ &+ \sum_{t\in\Delta:\ \mathrm{sl}} |A_t(\{l_e\})|\delta_t^{\mathrm{L},\pm}(\{l_e\})\,, \end{aligned} \tag{1}$$

with $|A_t|$ the absolute value of the triangle area. In the presence of a boundary, $\partial\Delta \neq \emptyset$, S_{R} attains boundary terms such that overall additivity is ensured. The equations of motion are given by $\frac{\partial S_{\mathrm{R}}}{\partial l_e} = 0$ supplemented by the Schläfli identity, $\sum_t |A_t|\frac{\partial \delta_t}{\partial l_e} = 0$.

Regge calculus can be formulated in terms of different variables [5, 6], with area variables [7–9] playing a particularly important role here due to their connection to LQG and spin-foams. The relation between area and length variables is in general not invertible, and area variables need to be further constrained to yield the dynamical equations of length Regge calculus [3, 6, 9]. However, in the symmetry reduced setting introduced below, the relation of area and length is globally invertible and the aforementioned constraints trivialize. For the remainder, we study the classical dynamics of area Regge calculus which will be used in future work as a comparison to results of the effective spin-foam path integral.

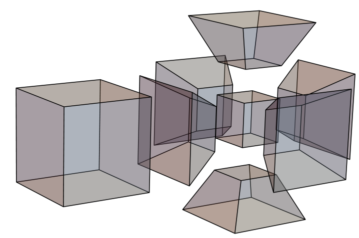

FIG. 1: Boundary of a 4-frustum given by two 3-cubes and six equal 3-frusta.

Lorentzian 4-frusta

Lorentzian 4-dimensional frusta are ideal to capture spatially flat, homogeneous and isotropic discrete geometries [10, 11]. The boundary of 4-frusta, depicted in Fig. 1, contains two spacelike 3-cubes and six 3-frusta that are either spacelike or timelike. Its entire geometry is captured by the area of squares contained in the 3-cubes, j_0 and j_1, and the area of the trapezoids contained in the 3-frusta, k. The general notions of Regge calculus introduced above are straightforwardly adapted to this setting. Gluing 4-frusta along boundary 3-frusta in spatial direction and along boundary 3-cubes in temporal direction, one obtains an extended cosmological spacetime of hypercubical combinatorics $\mathcal{X}_{\mathcal{V}}^{(4)}$ with slices labelled by $n \in \{0, ..., \mathcal{V}_T\}$ discretized by $\mathcal{V}_S^3$ spacelike 3-cubes. We say that a 4-frustum lies in a "slab" between slices n and $n+1$.

The squared volume of subcells connecting neighboring slices, i.e. 3-frusta, trapezoids and edges, which are detailed in Appendix **??**, carries information on the causal structure. If it is positive (negative), the corresponding subcell is timelike (spacelike). In the symmetry reduced setting, the ratio $\frac{|j_n - j_{n+1}|}{4k_n}$ fully characterizes the causal character of subcells suggesting separating the 4-frusta configurations into three sectors summarized in Tab. **??**.

As discussed in [11–15], two of these sectors I and II exhibit an irregular light cone structure, also referred to

[1] For the remainder, we neglect lightlike edges, triangles and tetrahedra. See [2] for the definition of deficit angles associated to $(d-1)$-dimensional null cells.

as causality violations, located at spacelike sub-cells. Examples of such irregularities are given by the trouser singularity and the yarmulke singularity associated to spatial topology change [16–18]. For causality violations at $(d-2)$-dimensional spacelike hinges the imaginary parts of the dihedral angles do not sum up to $\mp i2\pi$, leading to a complex valued Regge action. Choosing "+" in the convention of Lorentzian angles, the amplitude $\mathrm{e}^{iS_{\mathrm{R}}}$ is exponentially suppressed for trouser and exponentially enhanced for yarmulke configurations, vice versa for "−" [3].Without a complexification of the variables, the Regge equations of motion for causality violating configurations are complex equations with real variables and therefore overconstrained. Moreover, the results of [11, 12] show that in the respective settings, no real solutions exist. Since this applies also to the setting used in this work we henceforth focus on the causally regular sector III.

Lorentzian Regge action for spatially flat cosmology

For a single 4-frustum between, say slice n and $n+1$, the causally regular Regge action is given by

$$S_{\mathrm{III}}^{(n)} = 6(j_n - j_{n+1})\sinh^{-1}\left(\frac{j_{n+1}-j_n}{\sqrt{16k_n^2+(j_n-j_{n+1})^2}}\right) + 12k_n\left[\frac{\pi}{2} - \cos^{-1}\left(\frac{(j_n-j_{n+1})^2}{16k_n^2+(j_n-j_{n+1})^2}\right)\right] - \Lambda V^{(4)}\,, \qquad (2)$$

where Λ is a cosmological constant and $V^{(4)}$ is the 4-volume defined in Eq. (**??**) as an equation of (j_n, j_{n+1}, k_n). By construction, the 4-frustum action is additive such that on a discretization $\mathcal{X}_{\mathcal{V}}^{(4)}$ with $\mathcal{V}_S^3$ vertices in spatial direction and $\mathcal{V}_T+1$ spatial slices, the action is given by $S_{\mathrm{R}} = \mathcal{V}_S^3\sum_{n=0}^{\mathcal{V}_T-1} S_{\mathrm{R}}^{(n)}$. Additivity of the action per slab implies that the equations of motion for a given geometric quantity, being either j_n or k_n, will only depend on neighboring geometric labels.

In the following, we study the discrete dynamics of spatially flat cosmology as governed by the Regge equations. To that end, we consider first the case of a single time step, $\mathcal{V}_T = 1$, in Sec. and generalize then to an arbitrary number of time steps in Sec. **??**.

RECAP: ONE TIME STEP

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CONCLUSION

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Appendix

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